WEIHUA DU

@ weihuad@cs.cmu.edu (+1) 347 251 2777 / (+86) 180 6877 8796 GHC 6711, Carnegie Mellon University, Pittsburgh, PA, US
https://stiglidu.github.io StigLidu https://scholar.google.com/citations?user=6CqQ4L8AAAAJ

EDUCATION

Ph.D. Student in Language Technologies Institute

Carnegie Mellon University

Aug. 2024 – Ongoing Pittsburgh, US

- Co-advised by Prof. Yiming Yang and Prof. Sean Welleck.
- Working on LLM agents, including improving robustness and capacity of agents, designing effective inference-time algorithms, and developing methods for long-context management towards complex tasks.

Bachelor of Engineering in Computer Science (Yao Class)

Tsinghua University

Sep. 2020 – Jun. 2024 Beijing, China

- GPA 3.94 / 4.00;
- Gold medalist at the 2018 China National Olympiad in Informatics (CNOI).

Visiting Student

Massachusetts Institute of Technology

Feb. 2023 – Aug. 2023 Cambridge, US

- Advised by Prof. Joshua B. Tenenbaum and Prof. Chuang Gan.
- Specializing in LLM Agents and Embodied AI.

RESEARCH EXPERIENCE

Research Intern

MIT-IBM Computing Research Lab

May. 2026 – Aug. 2026 Cambridge, US

- Advised by Dr. Danish Contractor and Dr. Dan Gutfreund;
- **Research:** Long-Horizon Agent (both harness development and model training).

Research Intern

ByteDance Seed

May. 2025 – Aug. 2025 San Jose, US

- Advised by Dr. Jiecao Chen;
- **Research:** Reinforcement training for LLM agents and long-context management [10, 11].

Research Intern

Shanghai Artificial Intelligence Laboratory

Aug. 2023 – Jul. 2024 Shanghai, China

- Advised by Dr. Wenwei Zhang and Dr. Kai Chen;
- **Research:** Iteration of InternLM, focusing on improving tool calling capabilities [5].

Research Intern

Institute for Interdisciplinary Information Sciences (IIIS)

Jun. 2022 – Jan. 2023 Beijing, China

- Advised by Prof. Yi Wu;
- **Research:** Reinforcement learning application for the building industry [1] and an RL algorithm for policy diversity [2].

PUBLICATION (SORT BY PUBLICATION DATE IN DESCENDING ORDER. *DENOTES EQUAL CONTRIBUTION)

[16] OdysSim: Building Foundation Models for Human Behavior Simulation [link] NeurIPS 2026

Authors: Xuhui Zhou*, Weiwei Sun*, **Weihua Du**, Jiarui Liu, Haojia Sun, Qianou Ma, Tongshuang Wu, Yiming Yang, Maarten Sap

[15] Why Search When You Can Transfer? Amortized Agentic Workflow Design from Structural Priors [link]EMNLP 2026

Authors: Shiyi Du, Jiayuan Liu, **Weihua Du**, Yue Huang, Jiayi Li, Yingtao Luo, Xiangliang Zhang, Vincent Conitzer, Carl Kingsford

[14] Mind the Sim2Real Gap in User Simulation for Agentic Tasks [link] COLM 2026

Authors: Xuhui Zhou*, Weiwei Sun*, Qianou Ma, Yiqing Xie, Jiarui Liu, **Weihua Du**, Sean Welleck, Yiming Yang, Graham Neubig, Sherry Tongshuang Wu, Maarten Sap

[13] GradAlign: Gradient-Aligned Data Selection for LLM Reinforcement Learning [link] COLM 2026

Authors: Ningyuan Yang*, **Weihua Du***, Weiwei Sun, Sean Welleck, Yiming Yang

[12] Training Proactive and Personalized LLM Agents [link] COLM 2026

Authors: Weiwei Sun, Xuhui Zhou, **Weihua Du**, Xingyao Wang, Sean Welleck, Graham Neubig, Maarten Sap, Yiming Yang

[11] Beyond the Context Window: Scaling Agentic RL via End-to-end Optimized Context Compression [link] ACL 2026

Authors: Miao Lu, Weiwei Sun, **Weihua Du**, Zhan Ling, Xuesong Yao, Kang Liu, Jiecao Chen

[10] Generalizable End-to-End Tool-Use RL with Synthetic CodeGym [link] ICLR 2026

Authors: **Weihua Du**, Hailei Gong, Zhan Ling, Kang Liu, Lingfeng Shen, Xuesong Yao, Yufei Xu, Dingyuan Shi, Yiming Yang, Jiecao Chen

[9] ZeroTuning: Unlocking the Initial Token's Power to Enhance Large Language Models Without Training [link]ICLR 2026

Authors: Feijiang Han, Xiaodong Yu, Jianheng Tang, Delip Rao, **Weihua Du**, Lyle Ungar

[8] Agentic-R1: Distilled Dual-Strategy Reasoning [link] EMNLP 2025

Authors: **Weihua Du**, Pranjal Aggarwal, Sean Welleck, Yiming Yang

[7] Optimizing Temperature for Language Models with Multi-Sample Inference [link] ICML 2025

Authors: **Weihua Du**, Yiming Yang, Sean Welleck

[6] Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge [link] NeurIPS D&B 2024

Authors: **Weihua Du***, Qiushi Lyu*, Jiaming Shan, Zhenting Qi, Hongxin Zhang, Sunli Chen, Andi Peng, Tianmin Shu, Kwonjoon Lee, Behzad Dariush, Chuang Gan

[5] T-Eval: Evaluating the Tool Utilization Capability of Large Language Models Step by Step [link] ACL 2024

Authors: Zehui Chen*, **Weihua Du***, Wenwei Zhang*, Kuikun Liu, Jiangning Liu, Miao Zheng, Jingming Zhuo, Songyang Zhang, Dahua Lin, Kai Chen, Feng Zhao

[4] Building Cooperative Embodied Agents Modularly with Large Language Models [link] ICLR 2024

Authors: Hongxin Zhang*, **Weihua Du***, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, Chuang Gan

[3] HAZARD Challenge: Embodied Decision Making in Dynamically Changing Environments [link] ICLR 2024

Authors: Qinhong Zhou*, Sunli Chen*, Yisong Wang, Haozhe Xu, **Weihua Du**, Hongxin Zhang, Yilun Du, Joshua B. Tenenbaum, Chuang Gan

[2] Iteratively Learn Diverse Strategies with State Distance Information [link] NeurIPS 2023

Authors: Wei Fu, **Weihua Du**, Jingwei Li, Sunli Chen, Jingzhao Zhang, Yi Wu

[1] Automatic Truss Design with Reinforcement Learning [link] IJCAI 2023

Authors: **Weihua Du***, Jinglun Zhao*, Chao Yu, Xingcheng Yao, Zimeng Song, Siyang Wu, Ruifeng Luo, Zhiyuan Liu, Xianzhong Zhao, Yi Wu

PREPRINTS & WORKSHOP PAPERS

[P4] HarnessEval-W: Agentifying the Evaluation of Visual Worlds [link]

Authors: Weiliang Chen, Haowen Sun, Jun Gao, Jiawei Chi, Hanyang Wang, et al. (43 authors, including **Weihua Du**)

[P3] Reinforcing Human Behavior Simulation via Verbal Feedback [link]

Authors: Weiwei Sun*, Xuhui Zhou*, Jiarui Liu, **Weihua Du**, Haojia Sun, Yiqing Xie, Qianou Ma, Sihao Chen, Mengting Wan, Longqi Yang, Pei Zhou, Sherry Wu, Sean Welleck, Graham Neubig, Yiming Yang, Maarten Sap

[P2] The Consensus Trap: Rescuing Multi-Agent LLMs from Adversarial Majorities via Token-Level Collaboration [link]

Authors: Jiayuan Liu, Shiyi Du, **Weihua Du**, Mingyu Guo, Vincent Conitzer

[P1] AdaExplore: Failure-Driven Adaptation and Diversity-Preserving Search for Efficient Kernel Generation [link] LLA@COLM 2026

Authors: **Weihua Du**, Jingming Zhuo, Yixin Dong, Andre Wang He, Weiwei Sun, Zeyu Zheng, Manupa Karunaratne, Ivan Fox, Tim Dettmers, Tianqi Chen, Yiming Yang, Sean Welleck

SELECTED PRIZES

Outstanding Graduate of Yao Class | Top 20% graduates in Yao Class — Jun. 2024

SenseTime Scholarship | Top 30 undergraduates in AI within China — Dec. 2023

Zheng Geru Scholarship | Comprehensive Excellent Award in Tsinghua University, top 20% — Oct. 2023

Andrew C. Yao Award (Recognition Prize) | Scholarship in Yao Class, top 20% — Sep. 2023

Mr. and Mrs. Qu Yuzhi Scholarship | Academic & Sport Award in Tsinghua University — Oct. 2022

China Collegiate Programming Contest (CCPC), Weihai Site | Rank 4, **Gold Medal** — Nov. 2021

Mr. and Mrs. Huang Yicong Scholarship | Comprehensive Excellent Award in Tsinghua University, top 20% — Oct. 2021

China National Olympiad in Informatics (CNOI) 2018 | Rank 32, **Gold Medal** — Aug. 2018

SERVICE

- **Reviewer**: ICML 2025/2026, NeurIPS 2025, SCALR@COLM 2025, AAAI 2026, ICLR 2026, COLM 2026, TMLR.
- **Student Organizing Committee**: LTI-SRS 2025
- **Teaching**:
  - Teaching Assistant for **Advanced NLP (CMU 11-711)**, Spring 2026. [Course]
  - Guest Lecturer on **LLM Reasoning** at Generative AI (UMass Amherst COMPSCI 690L), Spring 2026. [Course]
- **Student Supervision**:
  - Qiushi Lyu (2024, Undergraduate at PKU; now Ph.D. at UIUC)
  - Ningyuan Yang (2025, Undergraduate at THU)
  - Yinjie Wang (2025, Ph.D. at UChicago)
  - Shuaiwei Huang (2026, Master at CMU)
  - Allen Wu (2026, Undergraduate at CMU)
- **Volunteer**:
  - Student Development Center of Tsinghua University. (187 hrs documented volunteer service)
  - President at CMU Recreational Badminton Club. (1000+ club members)